REVISION WORKSHEET

UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

AL KHALIDIYA BOYS SCHOOL

Cycle 3 (10048) | Dibba Al Hisn | Sharjah Eastern

Interference Practice Questions

Grade 10 Advanced | Module 17 Lesson 1 | Interference

Name: _____ Class: _____ Date: _____

Answer all 30 questions. Select one option for each MCQ. Show equations, working, units and explanations for written questions.

Useful equations and guidance: Double slit: Δx = λL/d; xm = mλL/d. Constructive path difference = mλ; destructive = (m + ½)λ. Small-angle approximation applies.

Term 1 2026–2027 | Textbook alignment: uploaded Grade 10 book, printed pages 41–49.

P01. Coherent sources maintain:

A. Unrelated frequencies and phases B. A constant phase difference
C. Zero wavelength D. Zero intensity at all times

P02. Constructive interference occurs when two equal-frequency waves arrive:

A. Always out of phase by half a cycle B. Without overlapping
C. In phase D. At different unrelated frequencies only

P03. For equal-amplitude waves, destructive interference is strongest at phase difference:

A. 0° B. 360°
C. 720° D. 180°

P04. In a double-slit experiment, increasing wavelength with other quantities fixed makes bright-fringe spacing:

A. Larger B. Smaller
C. Zero D. Unchanged

P05. A central bright fringe in the ideal symmetric double-slit pattern corresponds to path difference:

A. One half wavelength B. Zero
C. An arbitrary nonzero distance D. One quarter wavelength always

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P06. In a double-slit experiment, λ = 500 nm, slit separation d = 0.250 mm and screen distance L = 2 m. Find adjacent bright-fringe spacing in millimetres.

A. 8 mm B. 2 mm
C. 4 mm D. -4 mm

P07. Double-slit fringe spacing is 4 mm at L = 2.4 m with d = 0.300 mm. Find the wavelength in nanometres.

A. 1000 nm B. 250 nm
C. -500 nm D. 500 nm

P08. For λ = 600 nm, d = 0.300 mm and L = 1.00 m, find the distance of bright fringe order m = 2 from the central maximum, in millimetres.

A. 4 mm B. 8 mm
C. 2 mm D. -4 mm

P09. For wavelength 500 nm, calculate the path difference at the bright fringe of order m = 2.

A. 2000 nm B. 1000 nm
C. 500 nm D. -1000 nm

P10. For a film with one phase reversal, the minimum nonzero thickness for constructive reflection at normal incidence is t = λ/(4n). Find t for λ = 500 nm and n = 1.50.

A. 166.7 nm B. 41.67 nm
C. 83.33 nm D. -83.33 nm

P11. Define interference.

P12. Define coherent light.

P13. What is a bright interference fringe?

P14. Explain why ordinary independent lamps do not produce a stable visible fringe pattern easily.

P15. Why can a soap film display colours in white light?

P16. In a double-slit experiment, λ = 500 nm, slit separation d = 0.250 mm and screen distance L = 2.5 m. Find adjacent bright-fringe spacing in millimetres.

P17. In a double-slit experiment, λ = 500 nm, slit separation d = 0.250 mm and screen distance L = 3 m. Find adjacent bright-fringe spacing in millimetres.

P18. Double-slit fringe spacing is 6 mm at L = 3.6 m with d = 0.300 mm. Find the wavelength in nanometres.

P19. Double-slit fringe spacing is 8 mm at L = 4.8 m with d = 0.300 mm. Find the wavelength in nanometres.

P20. For λ = 600 nm, d = 0.300 mm and L = 1.00 m, find the distance of bright fringe order m = 3 from the central maximum, in millimetres.

P21. For λ = 600 nm, d = 0.300 mm and L = 1.00 m, find the distance of bright fringe order m = 4 from the central maximum, in millimetres.

P22. For wavelength 500 nm, calculate the path difference at the bright fringe of order m = 3.

P23. For wavelength 500 nm, calculate the path difference at the bright fringe of order m = 4.

P24. For a film with one phase reversal, the minimum nonzero thickness for constructive reflection at normal incidence is t = λ/(4n). Find t for λ = 550 nm and n = 1.50.

P25. For a film with one phase reversal, the minimum nonzero thickness for constructive reflection at normal incidence is $t = \lambda/(4n)$. Find t for $\lambda = 600$ nm and $n = 1.50$.

P26. In a double-slit experiment, predict the fringe-spacing change if screen distance doubles while all else stays fixed.

P27. If slit separation doubles with the same wavelength and screen distance, what happens to fringe spacing?

P28. Why does a white-light double-slit pattern have a white central fringe and coloured neighbouring fringes?

P29. A student applies the one-phase-reversal thin-film formula without checking surrounding media. Evaluate the procedure.

P30. Design a method to determine wavelength with a double slit.